

WaRF

When AI Reviewers Fight

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One question

Imagine five senior developers
with different specialties
reviewing your pull request.

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stop asking for more opinions?

"After two."

"Depends who the two are."

Exactly. That's WaRF.

Now replace those developers with five AI reviewers.

The problem

Static analyzer

- ✗ False positive
- ✗ False positive
- ✗ False positive

Developers stop listening.

One LLM reviewer

???

*How much
should I trust it?*

Five LLM reviewers

- ✓ APPROVE
- ✗ REJECT
- ✓ APPROVE
- ✗ REJECT
- ✓ APPROVE

Who should I trust?

Five reviewers. One decision.

AUDITOR	compliance
BOUNDARY	edge cases
ADVOCATE	intent
SKEPTIC	adversarial
TEST-FOCUSED	coverage



ONE DECISION

But WaRF doesn't treat
their opinions equally.

+ BANDIT (SAST, $\omega=0.75$, SECURITY only) — deterministic, no persona, available via `--sast`.

A 2500-year-old bug report

“Wolf!”

A shepherd boy raises the alarm so many times that the village stops listening.

The day a real wolf comes, his warning is correct —
and unanimously ignored.

Aesop’s point: credibility is not about volume of agreement, but about who was right.

A 2 500-year-old bug report

"Wolf!"

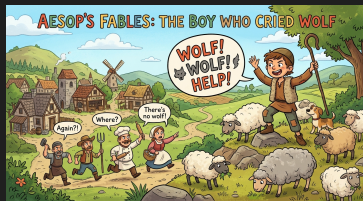
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WaRF's question: what if the person shouting

"Wolf!" is usually right — and the majority keeps outvoting them anyway?



Not all opinions are equal

Majority voting

5 reviewers

4 approve, 1 rejects



APPROVE

*What if the person
shouting "Wolf!"
is usually right?*

Not all opinions are equal

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4 approve, 1 rejects

↓

APPROVE

*What if the person
shouting “Wolf!”
is usually right?*

WaRF

AUDITOR

historically reliable in this domain

TEST-FOCUSED

historically unreliable in this domain

Everyone gets a vote.

Not every vote gets
the same weight.

The agents reason about the code. WaRF reasons about the agents.

ω is not assigned — estimated from oracle-labelled history, separately per domain.

**Let's see what that
actually looks like.**

What just happened?

Agent	Vote
AUDITOR	REJECT
ADVOCATE	REJECT
BOUNDARY	REJECT ← threshold crossed
SKEPTIC	<i>skipped</i>
TEST-FOCUSED	<i>skipped</i>

Each vote updates a running evidence score $\Lambda = \sum_i \omega_i \cdot \Delta_i$.

WaRF stops when Λ crosses a predefined boundary.

WaRF does not count votes.

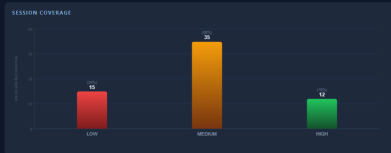
It accumulates evidence, weighted by each reviewer's track record, until a decision threshold is crossed.

The exact trace from this session is shown alongside.

Does WaRF know when it's right?

WaRF — Reliability by Confidence Band

Does WaRF's confidence predict its accuracy? Dashed line = random baseline (50%). n = oracle-labelled sessions per band.



In the calibration corpus, higher confidence coincides with higher accuracy.

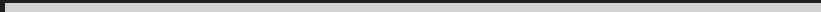
Preliminary: calibration and evaluation currently overlap.

Exact rates and session counts →

What early stopping buys

FULL REVIEW — all 5 agents queried

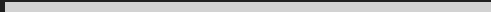
AUDITOR → ADVOCATE → BOUNDARY → SKEPTIC → TEST-FOCUSED



← tokens spent

EARLY STOP — SPRT boundary crossed at agent 3

AUDITOR → ADVOCATE → BOUNDARY → ✓ DECISION

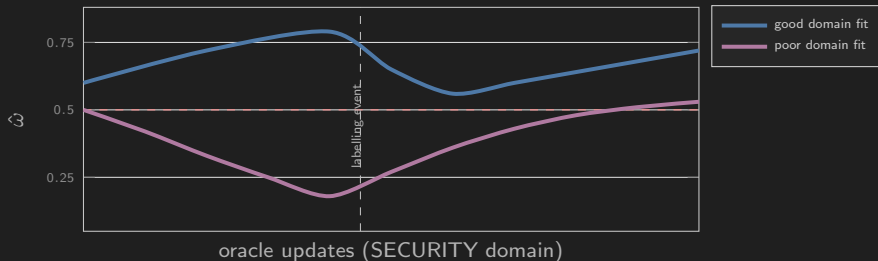


← tokens spent

Fewer agent calls are a consequence of sufficient evidence — not the optimisation target.

Observed token savings and stopping rates →

It actually learns



Case studies (SECURITY domain):

AUDITOR & ADVOCATE approved a urllib3 TLS change;
upstream reverted it, oracle said REJECT — their SECURITY weights drop.

TEST-FOCUSED over-rejects mature library code.

WaRF learns to trust it differently in that context.

Weights are not measures of quality — they are estimates of agent–domain fit.

Live weight trajectories for all agents →

One more.

This time the agents disagree.

Can Debate Resolve Disagreement?

Case	Outcome	Why
Authentication bug	Fixed	Correct minority had a verifiable argument
False belief	No change	Correct explanation did not penetrate
Plausible misconception	Worse	Wrong argument persuaded the correct minority

Debate can improve a decision. It can also amplify an error.

"Do you revise?" → convergence pressure

"Maintain or revise?" → minority keeps its footing

In our experiments, even prompt wording changed group behaviour.

Persuasion $\neq$ correctness.

When WaRF Fails

Failure mode	What happens
Persona bias	Two personas share a framing blind spot — AUDITOR and ADVOCATE read intent, miss the omission
Knowledge gap	Credential leak in requests: nobody recognises the pattern — even the one REJECT cites other reasons
False consensus	Correct minority capitulates in debate — persuasive wrong argument wins

More agents of the same kind make knowledge-gap failures worse.

WaRF cannot manufacture knowledge that none of its agents possess.

Everyone gets a vote.

Not every vote gets the same weight.

What's next

- More history → better calibration → less manual labelling
- More domains, more agent personas
- Open: should the oracle reward *reasoning quality*, not just vote correctness?

Prior art: CONSOL (2025, arXiv 2503.17587) applies SPRT to a single model at varied temperature. A ConSol-style pool inside WaRF — SKEPTIC×5 — rejected all 4 files it was given, each at HIGH confidence, including clean code (`hooks.py`, `sessions.py`): five identical pessimistic personas cannot say APPROVE. The mixed panel approved `hooks.py` because ADVOCATE and AUDITOR pushed back. Rerun on a balanced 12-file set with today's personas: accuracy is a tie (mixed panel 11/12, five AUDITORS 10/12, five SKEPTICS 10/12) — but only the clones were confidently wrong: five SKEPTICS unanimously rejected clean code at HIGH confidence.

Temperature diversity gives you noise around one perspective. Persona diversity gives you genuinely different expertise.

Why not two well-calibrated agents?

For any pool of n agents, the maximum signal that can accumulate in one direction is:

$$|\Lambda|_{\max}(n) = \left(\sum_{i=1}^n \omega_i \right) \times LLR_{ABS}$$

Early stopping requires $|\Lambda|_{\max}$ to reach a boundary: $|\lambda_B| \approx 2.25$ (REJECT) or $\lambda_A \approx 2.89$ (APPROVE).

At current ω values, the best two agents fall **below both boundaries** — no early stop, never HIGH.

Adding a third agent crosses the REJECT boundary.

Current weights & boundary chart $\rightarrow$

Counter to arXiv 2604.02460: code review $\neq$ reasoning task. Ensemble value is in HIGH-confidence early stops, not average accuracy.

A fraction of 5-agent sessions fire early with **High** confidence.

Structurally impossible with $n = 2$.

On those sessions:

5 agents is **cheaper** (3 used) and **more confident** than 2 agents (never HIGH).